Chapter 8. Causal Data Science

Transportability, selection bias, data fusion, missing data

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Motivation. The central question is how to answer a causal query about a target population when the available data sets were collected under different experimental conditions, in different populations, with non-random sampling, with missing entries, or measuring different variables.

8.1 Motivation and big picture

Motivation. Explain why real data rarely match the inferential target and how the mismatches are organized.

Examples. *Genetics:* experiments found a substance carcinogenic in rats; is it carcinogenic in humans? *Advertisement:* ML experts estimated the effect of advertisement strategies on sales of product A; what can be borrowed to advertise product B? *Robotics:* a planetary rover trained in the California desert to dig rocks; how can it use what it learned on Earth to minimize costly exploration on Mars?

"Not all data are created equal." Data are abundant but almost invariably collected under conditions that differ from the target:

(1) different experimental conditions, (2) different populations, (3) non-random sampling, (4) non-random treatment assignment, (5) unmeasured relevant variables. The collected data are messy and rarely match the inferential target. The good news: these dimensions have been formalized, with conditions and algorithms.

Heterogeneous data sets. For a target $Q = P(y \mid \text{do}(x))$, data sets may differ along four dimensions:

		Dataset 1	Dataset 2	Dataset n
d_1	population	Los Angeles	New York	Seoul
d_2	obs./exp.	experimental (randomized Z_1)	observational	experimental (randomized Z_2)
d_3	sampling	selection on age	selection on SES	–
d_4	measured	$\{X_1, Z_1, W, M, Y_1\}$	$\{X_1, X_2, Z_1, N, Y_2\}$	$\{X_2, Z_1, W, L, M, Y_1\}$

Goal. Data science = making the most of multiple, big and heterogeneous data sets: a principled way of performing inferences common in AI, ML and the empirical sciences, with formal machinery (language, conditions, algorithms) for causal inference and *fusion* across heterogeneous studies; imagine an automated scientist.

Classic identifiability takes (1) a query $P_x(y) = P(y \mid \text{do}(x))$, (2) a causal diagram, (3) data $P(\mathbf{V})$, and an inference engine returns either "no" (with evidence) or a formula $P_x(y) = f_{\mathcal{G}}(P(\mathbf{V}))$. In data fusion the target may be the US while data come from NY, LA, Boston, SF, Texas, Arkansas, Utah, Wyoming, each with its own diagram ($\mathcal{G}_{US}, \mathcal{G}_{NYC}, \dots$) and distribution ($P_{NYC} + P_{LA} + \cdots$). The sources differ in many ways:

(a) US: census data available (the target population);

(b) NY: survey data resembling the target;

(c) LA: survey data, younger subjects;

(d) Boston: age not recorded, mostly successful lawyers;

(e) SF: high post-treatment blood pressure;

(f) Texas: mostly Spanish subjects, high attrition;

(g) Arkansas: randomized trial on college students;

(h) Utah: RCT with paid volunteers, mainly unemployed;

(i) Wyoming: natural experiment on young athletes.

Tasks. Writing each data collection as a tuple (d_1, d_2, d_3, d_4) (population, obs./exp., sampling, measurement):

1. Descriptive statistics: $(d_1, \text{Samples}(\text{Obs}), d_3, d_4) \to (d_1, \text{Distribution}(\text{Obs}), d_3, d_4)$.
2. Experimental statistics: $(d_1, \text{Samples}(\text{do}(x)), d_3, d_4) \to (d_1, \text{Distribution}(\text{do}(x)), d_3, d_4)$.
3. Causal inference from observational studies: $(d_1, \text{Obs}, d_3, d_4) \to (d_1, \text{do}(x), d_3, d_4)$. "Causal inference" in 1974–2011 was mainly about confounding bias (Rubin, Robins, Dawid, Pearl, ...).
4. Generalized identification, i.e. using experiments on other variables: $(d_1, \text{do}(Z), d_3, d_4) \to (d_1, \text{do}(x), d_3, d_4)$.
5. Sampling selection bias: $(d_1, d_2, \text{Select}(\text{Age}), d_4) \to (d_1, d_2, \{\}, d_4)$.
6. Transportability (external validity): $(\text{Bonobos}, d_2, d_3, d_4) \to (\text{Humans}, d_2, d_3, d_4)$.

Each was addressed in isolation in the literature, usually under specific parametric conditions; in practice they appear together, which is what is called "causal data science".

Four dimensions. (1) Experimental conditions: generalized identification.

(2) Environmental conditions: transportability.

(3) Sampling conditions: recovering from selection bias.

(4) Responding conditions: recovering from missingness. Plus measurement (partial observation).

8.2 Experimental conditions: general identifiability

Motivation. Identify an effect from observational data combined with experiments on other variables.

8.2.1 The challenge

Diet Z, cholesterol X, heart attack Y, with $Z \to X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow Y$. Measured: observational $P(x, y, z)$ and experimental $P(x, y \mid \text{do}(z))$. Needed:

$$Q = P(y \mid \text{do}(x)) = \frac{P(x, y \mid \text{do}(z))}{P(x \mid \text{do}(z))} = P(y \mid \text{do}(z), x).$$

(Rule 3 adds $\text{do}(z)$ because Z affects Y only through X once its incoming edges are cut; Rule 2 exchanges $\text{do}(x)$ for x because in $\mathcal{G}_{\overline{Z}\underline{X}}$ there is no back-door path from X to Y.) Q is not identifiable from $P(x, y, z)$ alone (the bow $X \leftrightarrow Z \leftrightarrow Y$ makes X confounded with its child), so the experiment is essential.

8.2.2 Subtleties of z-ID

Three graphs with the same query (rows: $\mathcal{G}$ with $P(x, y, z)$; $\mathcal{G}_{\overline{Z}}$ with $P(x, y \mid \text{do}(z))$; $\mathcal{G}_{\overline{X}}$ for the query):

- $Z \to X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow Y$: **z-ID** (as above).
- $Z \to X \to Y$, $X \leftrightarrow Y$, $Z \leftrightarrow Y$: **not z-ID**; $X \leftrightarrow Y$ persists in $\mathcal{G}_{\overline{Z}}$, so the experiment on Z still leaves X confounded with Y. (Linear IV would help, nonparametric IV does not.)
- $Z \leftrightarrow X$, $Z \to Y$, $Z \leftrightarrow Y$, $X \to Y$: **not z-ID**; Z does not cause X, so intervening on Z gives no leverage on X.

8.2.3 z-ID reduced to calculus

Graph: $Z \to X \to W \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow W$, $Z \leftrightarrow Y$, $X \leftrightarrow Y$, with $P(\mathbf{v})$ and $P(\mathbf{v} \mid \mathrm{do}(z))$ available.

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &= \sum_w P(y \mid \mathrm{do}(x), w) P(w \mid \mathrm{do}(x)) = \sum_w P(y \mid \mathrm{do}(x, w)) P(w \mid \mathrm{do}(x)) \\ &= \sum_w \underbrace{P(y \mid \mathrm{do}(w))}_{Q[Y]} \underbrace{P(w \mid \mathrm{do}(x))}_{Q[W]}. \end{aligned}$$

Neither factor is identifiable from $P(\mathbf{V})$, but both are from $P(\mathbf{v} \mid \mathrm{do}(z))$:

$$\begin{aligned} Q[Y] = P(y \mid \mathrm{do}(w)) &= P(y \mid \mathrm{do}(w, z)) = \sum_x P(y \mid \mathrm{do}(w, z), x) P(x \mid \mathrm{do}(w, z)) \\ &= \sum_x P(y \mid \mathrm{do}(w, z), x) P(x \mid \mathrm{do}(z)) = \sum_x P(y \mid \mathrm{do}(z), w, x) P(x \mid \mathrm{do}(z)), \\ Q[W] = P(w \mid \mathrm{do}(x)) &= P(w \mid \mathrm{do}(x, z)) = P(w \mid \mathrm{do}(z), x). \end{aligned}$$

Within the submodel $\mathcal{M}_z$, $Q[Y]$ is a front-door/back-door type computation and $Q[W]$ an unconfounded effect.

8.2.4 Example: drug–drug interactions

Y: cardiovascular disease, B: blood pressure, X_1: antihypertensive drug, X_2: anti-diabetic drug. Base graph (a): $X_1 \to B \to Y$, $X_2 \to Y$, $X_1 \leftrightarrow B$, $X_1 \leftrightarrow X_2$, $X_2 \leftrightarrow Y$. Goal: the effect of prescribing *both* drugs on cardiovascular risk, from individual-drug experiments, $\mathbb{Z} = \{\{X_1\}, \{X_2\}\}$:

$$P_{x_1,x_2}(y) = \sum_b P_{x_1,x_2}(y, b) = \sum_b P_{x_1}(b) P_{x_2,b}(y) = \sum_b P_{x_1}(b) P_{x_2}(y \mid b).$$

$P_{x_1}(b)$ comes from the X_1 experiment; $P_{x_2}(y \mid b)$ from the X_2 experiment (in $\mathcal{M}_{x_2}$, B is unconfounded with Y). Adding $B \leftrightarrow X_2$ (b) keeps identifiability; adding $B \leftrightarrow Y$ (c) or $X_1 \leftrightarrow Y$ (d) destroys it.

8.2.5 Algorithm and summary

A two-phase algorithm using probability axioms and do-calculus:

(1) the query is modified and decomposed into factors;

(2) each factor is identified from one of the available distributions $\mathbb{P} = \{P_{\mathbf{Z}_1}, \ldots, P_{\mathbf{Z}_m}\}$. It fails only if some factor cannot be identified from any available distribution.

The identifiability of any expression $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z})$ can be decided for any graph; if identifiable from a combination of observational and experimental studies, the estimand is derived in polynomial time; do-calculus is complete. References: Pearl (1993) back-door; Pearl (1994) do-calculus; Tian and Pearl (2002) c-components and algorithmic ID; Shpitser and Pearl, Huang and Valtorta (2006) completeness of ID; Bareinboim and Pearl (2012) completeness of z-ID; Lee, Correa and Bareinboim (2019) g-ID; Correa and Bareinboim (2019, 2020) stochastic interventions (σ-calculus).

8.3 Environmental conditions: transportability

Motivation. Transfer causal knowledge from source populations to a different target population.

8.3.1 The problem

Can the effect of $\mathbf{X}$ on $\mathbf{Y}$ in a target environment Π^* be computed from observational and experimental findings in different populations (e.g. applying US education policies to South Korea)? "Environment" can mean population, domain, etc. Other sciences call this external validity: "to what populations, settings, treatment variables and measurement variables can this effect be generalized?" (Shadish, Cook and Campbell 2002); extrapolation across studies requires "some understanding of the reasons for the differences" (Cox 1958); an experiment has external validity if the outcome distribution of the treatment group equals that of an actual program (Manski 2007). (Internal validity asks whether a study reliably establishes cause and effect in its own population.)

From lab to real world. At one extreme (H_0) all mechanisms are the same in source and target ($f_z = f_z^*$, $f_x = f_x^*$, ...): trivially transportable. At the other (H_a) everything differs: nothing transports. Real problems lie on the spectrum.

8.3.2 What can an experiment in LA tell us about NYC?

Z (age) $\to X$, $Z \to Y$, $X \to Y$; age distributions differ between LA and NYC, all mechanisms equal otherwise. LA: experimental study, $P(x, y, z)$ and $P(y \mid \mathrm{do}(x), z)$; NYC: observational, $P^*(x, y, z)$. Needed:

$$Q = P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z)\, P^*(z)\,:$$

the z-specific effects (LA) reweighted by the NYC age distribution.

Transport formulas depend on the causal story. (a) Z is age ($Z \to X$, $Z \to Y$, differing in distribution): $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P^*(z)$.

(b) Z is language skill, a proxy affected by the population difference but not causally relevant: $P^*(y \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x))$.

(c) Z is a bio-marker on the causal path $X \to Z \to Y$: $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P^*(z \mid x)$. ($P^*$: target, P: source.)

8.3.3 Selection diagrams

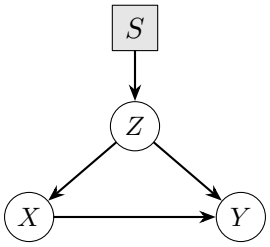


Figure. *A selection diagram for the LA/NYC example: the selection node S points into Z (age), marking the only mechanism that differs between the source and target populations. Since S affects Y only through Z, $P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P^*(z)$.*

Selection diagrams encode disparities and commonalities among domains. Overlay the source diagram $\mathcal{G}$ and target diagram $\mathcal{G}^*$ and add a *selection node* S (drawn as a small square) pointing to each variable whose mechanism (or exogenous distribution) may differ. Example: $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to W \to Y$; $\mathcal{G}^*$ lacks $Z \to X$. With $f_Z = f_Z^*$, $f_X \neq f_X^*$, $f_W = f_W^*$, $f_Y \neq f_Y^*$, the selection diagram has $S \to X$ and $S \to Y$. (If $Z \to X$ is absent in $\mathcal{G}$ but present in $\mathcal{G}^*$, the union graph is used.) Queries in the target are written $P^*(\cdot) = P(\cdot \mid S = s^*)$.

Theorem 8.1 (Transportability; Pearl and Bareinboim 2011)

Q is transportable from Π to Π^* iff there is a do-calculus reduction of $Q(\Pi^*)$ to an estimand that is a function of the observed distributions.

Proof [Supplement]. *If:* the selection diagram encodes both domains as one causal model with selection nodes S, and $Q(\Pi^*) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), S = s^*)$. Each do-calculus step is valid in this augmented model, so a reduction to an expression in the observable distributions (with S appearing only in factors estimable in Π^*) expresses $Q(\Pi^*)$ through available data. *Only if:* completeness of do-calculus for transportability is proved in Bareinboim and Pearl (2012) by constructing, for non-transportable queries, two pairs of domains that agree on all available distributions but not on $Q(\Pi^*)$; not reproduced here. ■

Example 8.1

Selection diagram with $S \to Z$, $Z \to X$, $Z \to W \to Y$, $X \to Y$, $Z \leftrightarrow X$, $Z \leftrightarrow W$, $X \leftrightarrow Y$:

$$\begin{aligned} Q = P^*(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x), S) = \sum_w P(y \mid \mathrm{do}(x), S, w) P(w \mid \mathrm{do}(x), S) \\ &= \sum_w P(y \mid \mathrm{do}(x), w) P(w \mid \mathrm{do}(x), S) = \sum_w P(y \mid \mathrm{do}(x), w) P(w \mid S) \\ &= \sum_w P(y \mid \mathrm{do}(x), w) P^*(w). \end{aligned}$$

Step 3 uses $(Y \perp_d S \mid X, W)_{\mathcal{G}_{\overline{X}}}$ (Rule 1); step 4 uses that X is not an ancestor of W (Rule 3). The first factor comes from the source experiment, the second from target observations.

Algorithm. Input: an annotated (selection) diagram. Output: whether Q is transportable, and if so a transport formula specifying which measurements to take in the source and which in the target, e.g.

$$P^*(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) \sum_w P^*(z \mid w) \sum_t P(w \mid \mathrm{do}(x), t) P^*(t).$$

Principle: any c-factor not identifiable from $P^*(\mathbf{V})$ must be identified from source experiments, which is possible only if no selection node points into the variables of that c-factor.

8.3.4 General transportability

Can we predict the effect of X on Y in Π^* using experiments over sets $\{\mathbf{Z}^1, \ldots, \mathbf{Z}^n, \mathbf{Z}^*\}$ from domains $\Pi = \{\Pi^1, \ldots, \Pi^n, \Pi^*\}$? Sometimes; the theory characterizes when and how.

Example 8.2

Transport $Q = P^*(y \mid \mathrm{do}(x))$ from Π^a (LA; experiments on X) and Π^b (NYC; experiments on Z). Both diagrams: $X \to Z \to Y$, $X \leftrightarrow Z$, $X \leftrightarrow Y$; in Π^a selection nodes point to X and Y; in Π^b to X and Z. Separately, Q is provably not transportable from either. Combined (writing S_{xzy} for the selection nodes),

$$\begin{aligned} Q = P(y \mid \mathrm{do}(x), S_{xzy}) &= \sum_z P(y \mid \mathrm{do}(x), z, S_{xzy}) P(z \mid \mathrm{do}(x), S_{xzy}) && \text{def., axioms} \\ &= \sum_z P(y \mid \mathrm{do}(x), \mathrm{do}(z), S_{xzy}) P(z \mid \mathrm{do}(x), S_{xzy}) && \text{Rule 2} \\ &= \sum_z P(y \mid \mathrm{do}(z), S_{xzy}) P(z \mid \mathrm{do}(x), S_{xzy}) && \text{Rule 3} \end{aligned}$$

$$= \sum_z P(y \mid \text{do}(z), S_y) P(z \mid \text{do}(x), S_z) \qquad \text{Rule 1}$$

$$= \sum_z \underbrace{P^{(b)}(y \mid \text{do}(z))}_{\text{NYC}} \underbrace{P^{(a)}(z \mid \text{do}(x))}_{\text{LA}} \qquad \text{def.}$$

($P(y \mid \text{do}(z))$ is needed where S does not point to Y, i.e. NYC; $P(z \mid \text{do}(x))$ where S does not point to Z, i.e. LA.) (Selection diagrams are defined for a pair of domains but generalize to multiple sources.)

8.3.5 Is the gold standard golden?

Before randomization: $Z \to X$, $Z \to Y$, $X \to Y$ with confounding and a selection node into Z. After randomization the edges into X disappear, but $S \to Z \to Y$ remains: the trial population differs from the target in Z. Even a perfect RCT requires a transportability exercise; transportability theory is unavoidable.

Summary. Nonparametric transportability can be decided when the problem is encoded in selection diagrams; transport formulas are derived in polynomial time; the calculus and the algorithm are complete. These results address long-standing problems of external validity, meta-analysis and heterogeneity. References: Pearl and Bareinboim (2011) semantics; Bareinboim and Pearl (2012) completeness; (2013a,b) multiple environments and limited experiments; (2014) completeness for those; Lee, Correa and Bareinboim (2020) subset experiments.

8.4 Sampling conditions: recovering from selection bias

Motivation. Recover population quantities from samples whose inclusion depends on the variables themselves.

8.4.1 The problem

Selection bias, caused by preferential inclusion (exclusion) of samples, is a major obstacle to valid causal and statistical inference. Augment the diagram with a selection indicator S ($S = 1$: in the sample). In Pearl's seven-variable model ($Z_1 \to W_1 \to X$, $Z_1 \to Z_3$, $Z_2 \to Z_3$, $Z_2 \to W_2 \to Y$, $Z_3 \to X$, $Z_3 \to Y$, $X \to W_3 \to Y$) an isolated S means random sampling; with $W_1 \to S$ and $W_2 \to S$, sampling depends on covariates. Under what conditions can we estimate, e.g., $P(y \mid x)$ from $P(\mathbf{v} \mid S = 1)$? The inference engine now receives selection-biased data.

8.4.2 The origin of selection bias

$X = U_x$, $Y = U_y$, $S = \neg(X \oplus Y)$, $P(u_x, u_y) = 1/4$. In the selected sample, $P(X = 1, Y = 1 \mid S = 1) = P(X = 0, Y = 0 \mid S = 1) = 1/2$, a strong correlation:

$$P(Y = 1 \mid X = 1, S = 1) - P(Y = 1 \mid X = 0, S = 1) = 1,$$

whereas in the population $P(X = x, Y = y) = 1/4$ and $P(Y = 1 \mid X = 1) - P(Y = 1 \mid X = 0) = 0$. Conditioning on a common effect ($X \to S \leftarrow Y$) creates dependence.

Confounding versus selection bias. Both are an unblockable flow of information between treatment and outcome (spurious correlation): confounding through a common cause $X \leftarrow Z \to Y$; selection through conditioning on a common effect $X \to S \leftarrow Y$.

Transportability versus selection. Nature-made versus man-made: in transportability S marks difference-producing factors across populations; in selection, S is the selection indicator and its parents are sample-producing factors.

8.4.3 Variants

In statistics:

(i) without external data, is the odds ratio $OR(y,x) = \frac{P(y|x)P(y'|x')}{P(y'|x)P(y|x')}$ recoverable from $P(\mathbf{v} \mid S=1)$? (ii) without external data, is $P(y \mid x)$ recoverable? (iii) with external data $P(\mathbf{t})$ for some $\mathbf{T} \subseteq \mathbf{V}$, is $P(y \mid x)$ recoverable? In causal inference: is $P(y \mid \text{do}(x))$ recoverable from $P(\mathbf{v} \mid S=1)$ and $P(\mathbf{t})$? With instruments, are bounds on $P(y \mid \text{do}(x))$ recoverable?

8.4.4 Recoverability without external data

Theorem 8.2 (Bareinboim, Tian and Pearl 2014)

$Q = P(y \mid x)$ is recoverable from selection-biased data iff $(S \perp\!\!\!\perp Y \mid X)$.

Proof [Supplement]. *If:* $S \perp\!\!\!\perp Y \mid X$ gives $P(y \mid x) = P(y \mid x, S=1)$, which is estimable from the selected sample. *Only if:* if $S \not\perp\!\!\!\perp Y \mid X$ in $\mathcal{G}$, Bareinboim, Tian and Pearl (2014) construct two models compatible with $\mathcal{G}$ that agree on $P(x, y \mid S=1)$ but differ in $P(y \mid x)$; the construction is not reproduced here. ■

If $Y \to S$, not recoverable; if only $X \to S$, recoverable since $P(y \mid x) = P(y \mid x, S=1)$.

8.4.5 Recoverability with external data

In the seven-variable model with $W_1 \to S \leftarrow W_2$, $P(y \mid x)$ is not recoverable by the previous theorem, but what if unbiased $P(W_1, W_2)$ is available?

Theorem 8.3 (Sufficiency)

$P(y \mid x)$ is recoverable if there is a set C with $(Y \perp\!\!\!\perp S \mid C, X)$ in $\mathcal{G}$ and $P(C, X)$ estimable; then

$$P(y \mid x) = \sum_c P(y \mid x, c, S=1) P(c \mid x).$$

Proof [Supplement]. By the law of total probability, $P(y \mid x) = \sum_c P(y \mid x, c) P(c \mid x)$. Since $(Y \perp\!\!\!\perp S \mid C, X)$, $P(y \mid x, c) = P(y \mid x, c, S=1)$, which is estimable from the selected sample, and $P(c \mid x)$ is available by assumption. Substituting gives the formula. ■

$C = \{W_1, W_2\}$: yes (the parents of S separate it from everything). $C = \{W_1, Z_1, Z_2\}$: no ($S \leftarrow W_2 \to Y$ is open). $C = \{W_2, Z_3\}$: yes.

Confounding and selection together. Examples with C given externally:

(1) $W_1 \to X$, $W_1 \to W_2 \to Y$, $X \to Y$, $W_2 \to S$: $C = \{W_2\}$ recovers $P(y \mid \text{do}(x))$; $C = \{W_1\}$ does not.

(2) $X \to Y$, $X \to W \leftarrow Y$, $W \to S$: $C = \{\}$ and $C = \{W\}$ fail, $C = \{W, X\}$ works.

Theorem 8.4 (Generalized adjustment; Correa, Tian and Bareinboim 2018)

$(\mathbf{Z}, \mathbf{Z}^T)$ with $\mathbf{Z}^T \subseteq \mathbf{Z}$ (the part measured without bias) is admissible for generalized adjustment relative to $(\mathbf{X}, \mathbf{Y})$ iff

1. $\mathbf{Z}$ does not disturb any proper causal path;
2. $\mathbf{Z} \cup \{S\}$ blocks all non-causal paths (cut the first edge of every proper causal path; then $\mathbf{Z} \cup \{S\}$ must separate $\mathbf{X}$ and $\mathbf{Y}$);
3. $\mathbf{Z}^T$ d-separates $\mathbf{Y}$ from S in the graph where proper causal paths have been severed.

Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} \underbrace{P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, S=1) P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T, S=1)}_{\text{biased data}} \underbrace{P(\mathbf{z}^T)}_{\text{unbiased data}} .$$

Proof [Supplement]. *Sufficiency.* By condition 3, $\mathbf{Y} \perp\!\!\!\perp S \mid \mathbf{X}, \mathbf{Z}$ in the relevant graph, so $P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, S=1)$. Conditions 1–2 make $\mathbf{Z}$ an adjustment set, so $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$. Finally $P(\mathbf{z}) = P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T) P(\mathbf{z}^T)$, where $P(\mathbf{z}^T)$ is measured without bias and the remaining condition makes $P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T) = P(\mathbf{z} \setminus \mathbf{z}^T \mid \mathbf{z}^T, S=1)$. *Necessity* is proved in Correa, Tian and Bareinboim (2018) and not reproduced here. ■

Example 8.3

With treatments $\mathbf{X}$, $\mathbf{Z} = \{Z_1, Z_2, Z_3\}$ satisfying the three conditions and $\mathbf{Z}^T = \{Z_2, Z_3\}$: $P(y \mid \mathrm{do}(\mathbf{x})) = \sum_{z_1, z_2, z_3} P(y \mid \mathbf{x}, z_1, z_2, z_3, S=1) P(z_1 \mid z_2, z_3, S=1) P(z_2, z_3)$.

Summary. Nonparametric recoverability can be decided given an augmented diagram with the selection mechanism; estimands are derived in polynomial time; results are complete for pure recoverability and sufficient with external information (Correa, Tian and Bareinboim 2019). The back-door criterion generalizes to handle confounding and selection simultaneously. References: Bareinboim, Tian and Pearl (2014) Markovian completeness; Bareinboim and Tian (2015) semi-Markovian; Correa, Tian and Bareinboim (2018) complete back-door; (2019a) ID with selection bias and external data; (2019b) complete adjustment for transportability with selection bias.

8.5 Responding conditions: missing data

Motivation. Use missingness graphs to decide when quantities are recoverable from incomplete data.

(Partly following Mohan and Pearl 2021, "Graphical models for processing missing data", JASA.)

8.5.1 Motivation

Sensors fail, respondents skip questions, patients cannot recall treatments or outcomes. The statistical literature, rooted in Rubin's work, mostly assumes Missing At Random (MAR): "essentially all the literature on multivariate incomplete data assumes MAR". Popular methods (maximum likelihood via EM, Dempster et al. 1977; multiple imputation, Rubin 1978) need MAR for consistency. But it is almost impossible to decide whether MAR holds, and the literature beyond MAR is limited. Graphical models, long used for missing data in Bayesian-network learning, encode the *reasons* for missingness.

Why graphs? *Transparency:* reasons for missingness are causal, not statistical; MCAR/MAR independences are invisible in the data but visible in the graph. *Recoverability:* decide from the model whether any method yields a consistent estimate of a parameter, and how. *Testability:*

decide whether the model's assumptions are compatible with the (corrupted) data. The graph implies conditional independences.

8.5.2 A motivating example

A school study measures Age A, Gender G, Obesity O (graph $G \to O \leftarrow A$). A and G are fully observed from school records; O is missing for students who do not reveal their weight, recorded as m. A proxy O^* is determined by O and the missingness mechanism R_O:

$$O^* = f(R_O, O) = \begin{cases} O & R_O = 0, \\ m & R_O = 1. \end{cases}$$

R_O masks ($R_O = 1$) or reveals ($R_O = 0$) obesity. Reasons for missingness: random (lost questionnaires); caused by a fully observed variable (rebellious teenagers, $A \to R_O$); caused by a partially observed variable (embarrassed obese students, $O \to R_O$).

8.5.3 m-graphs

Nodes are partitioned into $\mathbf{U} \cup \mathbf{V}_o \cup \mathbf{V}_m \cup \mathbf{V}^* \cup \mathbf{R}$: latent variables $\mathbf{U}$; variables observed in every record $\mathbf{V}_o$; variables missing in at least one record $\mathbf{V}_m$; proxies $\mathbf{V}^*$ (V_i^* is actually observed for $V_i \in \mathbf{V}_m$); missingness mechanisms $\mathbf{R}$. No variable in $\mathbf{V}_o \cup \mathbf{V}_m \cup \mathbf{U}$ is a child of an R variable unless stated otherwise. The *observed-data distribution* is $P(\mathbf{V}_o, \mathbf{V}^*, \mathbf{R})$; the *underlying distribution* (had there been no missingness) is $P(\mathbf{V}_o, \mathbf{V}_m, \mathbf{R})$.

8.5.4 Three categories

- **MCAR:** $\mathbf{V}_m, \mathbf{V}_o, \mathbf{U} \perp\!\!\!\perp \mathbf{R}$; no edges between $\mathbf{R}$ and $\mathbf{V}_o \cup \mathbf{V}_m$. Example: $O, A, G \perp\!\!\!\perp R_O$.
- **MAR:** $\mathbf{V}_m, \mathbf{U} \perp\!\!\!\perp \mathbf{R} \mid \mathbf{V}_o$;

  (i) no edge between any R and a partially observed variable, (ii) no bidirected edge between an R and a fully observed variable. Example: $A \to R_O$, $O \perp\!\!\!\perp R_O \mid A, G$.
- **MNAR:** neither. Example: $O \to R_O$, $O \not\perp\!\!\!\perp R_O \mid A, G$.

Recovering under MCAR. Since R_O is independent of the variables, $P(G, O, A) = P(G, O, A \mid R_O = 0)$, and $R_O = 0$ implies $O^* = O$, so $P(G, O, A) = P(G, O^*, A \mid R_O = 0)$: list-wise deletion (complete-case analysis), which usually wastes data.

Recovering under MAR. Factorize $P(G, O, A) = P(G, O \mid A)P(A)$. A d-separates (G, O) from R_O, so $= P(G, O \mid A, R_O = 0)P(A) = P(G, O^* \mid A, R_O = 0)P(A)$. Alternatively $P(G, O, A) = P(O^* \mid A, G, R_O = 0)P(A, G)$.

MCAR implies MAR, so MAR methods apply to MCAR. Even though list-wise deletion is unbiased under MCAR, compare the efficiency of $P(G, O^*, A \mid R_O = 0)$ (MCAR estimator) with $P(O^* \mid A, G, R_O = 0)P(A, G)$ (MAR estimator, using all records for $P(A, G)$).

Non-recoverability under MNAR. The edge $O \to R_O$ prevents transforming $P(G, O, A)$ into an expression in observables; in this example the joint distribution is provably non-recoverable.

A recoverable MNAR example. $Z_1 \to Z_2$, $Z_1 \to X$, $Z_1 \to Y$, $Z_2 \to X$, $X \to Y$ (and $Z_2 \to Y$), with Z_1 pointing into R_{Z_2}, R_X, R_Y, and $Z_2 \to R_{Z_2}$ etc. $P(Z_1)$ is recoverable since $R_{Z_1} \perp\!\!\!\perp Z_1$: $P(Z_1) = P(Z_1 \mid R_{Z_1} = 0) = P(Z_1^* \mid R_{Z_1} = 0)$, but *not* from complete cases, since $Z_1 \not\perp\!\!\!\perp R_X$, so $P(Z_1) \neq P(Z_1 \mid \mathbf{R} = 0)$. $P(Y \mid X, Z_1, Z_2)$ is recoverable from complete cases: $P(Y \mid X, Z_1, Z_2) = P(Y \mid X, Z_1, Z_2, \mathbf{R} = 0) = P(Y^* \mid X^*, Z_1^*, Z_2^*, \mathbf{R} = 0)$. (Proxies are often omitted from m-graphs for clarity.)

8.5.5 General results

Theorem 8.5 (Sequential factorization)

Given an m-graph and $P(\mathbf{V}^*, \mathbf{V}_o, \mathbf{R})$, a target Q is recoverable if it can be decomposed into an ordered (chain-rule) factorization, or a sum of such, in which every factor $Q_i = P(Y_i \mid X_i)$ satisfies $Y_i \perp\!\!\!\perp (R_{y_i}, R_{x_i}) \mid X_i$. Then each $Q_i = P(Y_i \mid X_i, R_{Y_i} = 0, R_{X_i} = 0)$.

Proof [Supplement]. Write $Q = \prod_i P(Y_i \mid X_i)$ (or a sum of such products). For each factor, $Y_i \perp\!\!\!\perp (R_{y_i}, R_{x_i}) \mid X_i$ gives $P(Y_i \mid X_i) = P(Y_i \mid X_i, R_{Y_i} = 0, R_{X_i} = 0)$. When the indicators are 0 the proxies equal the true values ($V^* = V$ if $R_V = 0$), so the right-hand side is $P(Y_i^* \mid X_i^*, R_{Y_i} = 0, R_{X_i} = 0)$, a functional of the observed distribution. Hence each factor, and Q, is recoverable. ■

Theorem 8.6 (R-factorization: recoverability of $P(\mathbf{V})$)

Given an m-graph with no edges between R variables, $P(\mathbf{V})$ is recoverable iff there is no $X \in \mathbf{V}_m$ such that (1) X and R_x are neighbors, or (2) X and R_x are connected by a path whose intermediate nodes are all colliders in $\mathbf{V}_m \cup \mathbf{V}_o$. When recoverable,

$$P(\mathbf{v}) = \frac{P(\mathbf{R} = 0, \mathbf{v})}{\prod_i P(R_i = 0 \mid \mathrm{Mb}^o_{r_i}, \mathrm{Mb}^m_{r_i}, R_{\mathrm{Mb}^m_{r_i}} = 0)},$$

where $\mathrm{Mb}^o_{r_i} \subseteq \mathbf{V}_o$ and $\mathrm{Mb}^m_{r_i} \subseteq \mathbf{V}_m$ form the Markov blanket of R_i.

Proof [Supplement]. *Formula, given the conditions.* By the chain rule, $P(\mathbf{v}) = P(\mathbf{v}, \mathbf{R} = 0)/P(\mathbf{R} = 0 \mid \mathbf{v})$. With no edges among the R variables, $P(\mathbf{R} = 0 \mid \mathbf{v}) = \prod_i P(R_i = 0 \mid \mathbf{v}, \mathbf{R}_{-i} = 0)$ factorizes over the Markov blankets of the R_i, and each factor equals $P(R_i = 0 \mid \mathrm{Mb}^o_{r_i}, \mathrm{Mb}^m_{r_i}, R_{\mathrm{Mb}^m_{r_i}} = 0)$. The two graphical conditions guarantee that the Markov blanket of R_i does not contain the variable whose missingness R_i indicates, so each factor involves only variables observed when the relevant indicators are 0; the numerator $P(\mathbf{v}, \mathbf{R} = 0)$ is observed directly. *Necessity* (non-recoverability when X and R_x are adjacent or joined by a collider path) is proved by Mohan and Pearl (2014, 2021) and not reproduced here. ■

8.5.6 Recovering a causal effect

Graph: $W \to Z \to Y$, $W \leftrightarrow Y$ and $R_Y \leftrightarrow W$, with Y partially observed ($Y \to Y^* \leftarrow R_Y$):

$$\begin{aligned} P(y \mid \mathrm{do}(z)) &= P(y \mid \mathrm{do}(z), r_y) = P(y^* \mid \mathrm{do}(z), r_y) \\ &= \sum_w P(y^* \mid w, \mathrm{do}(z), r_y) P(w \mid \mathrm{do}(z), r_y) = \sum_w P(y^* \mid w, z, r_y) P(w \mid r_y), \end{aligned}$$

with r_y denoting $R_Y = 0$.

Conclusion. Practising statisticians should model the process causing missing data far more often; to do so they need models of that process, which have received little attention in the statistical literature.

8.6 Measurement: general identifiability with partial observation

Motivation. Combine experiments that each measure only some variables.

General identifiability uses experiments $\mathbb{P} = \{P_{\mathbf{Z}_i}(\mathbf{V})\}_i$. But one cannot always measure all of $\mathbf{V}$: social scientists measure only variables relevant and accessible to their study; clinical trials on

different diseases monitor different variables (oximeter for COVID vs. carbon-monoxide poisoning). *General identifiability with partial observation* (GID-PO) uses $\mathbb{P} = \{P_{\mathbf{Z}_i}(\mathbf{V}_i)\}_i$ with $\mathbf{V}_i \subseteq \mathbf{V}$.

Example 8.4 (Exercise and stroke)

Age $A \to$ exercise X, $A \to$ BMI B, $X \to B$, $B \to$ blood pressure C, $C \to$ stroke Y, $X \leftrightarrow Y$. Task: the effect of X on Y, given an experimental study on exercise measuring age and blood pressure, $P_X(A, C)$, and an observational study measuring BMI, blood pressure and stroke, $P(B, C, Y)$:

$$
\begin{aligned}
P_x(y) &= \sum_{a,b,c} P_x(a,b,c,y) = \sum_{a,b,c} P(a)P_{a,x}(b)P_b(c)P_c(y) && \text{factorization} \\
&= \sum_{a,c} P(a)\Big[\sum_{b} P_{a,x}(b)P_b(c)\Big]P_c(y) = \sum_{a,c} P(a)P_{a,x}(c)P_c(y) && \text{proposition} \\
&= \sum_{a,c} P_x(a)P_x(c \mid a)\sum_{b} P(y \mid b,c)P(b) && \text{GID of each factor.}
\end{aligned}
$$

The bracketed factors merge into $P(a)P_{a,x}(c)$, identified from $P_X(A, C)$ (since A is not affected by X, $P(a) = P_x(a)$), while $P_c(y) = \sum_b P(y \mid b, c)P(b)$ is identified from $P(B, C, Y)$ by back-door adjustment for B (which blocks $C \leftarrow B \leftarrow X \leftrightarrow Y$). The pieces fit like a jigsaw puzzle.

8.7 Conclusions

- Principled use of abundant data requires representing the underlying causal mechanisms, typically in a coarse and parsimonious way.
- There are necessary and sufficient conditions and algorithms for the most common biases in data collection, solving long-standing challenges: confounding bias, selection bias and generalizability of experiments (sufficient conditions for missing data and partial measurement).
- This gives a principled framework for causal data science: pooling observational and experimental information and knowledge spread across multiple heterogeneous domains.